# Mirror Neuron Contributions to Social Synchrony and Collaborative Learning Across Species

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## Abstract

Mirror neurons activate both when a person performs an action and when they watch someone else performing the same action. Past research has shown that this system becomes more active during socially meaningful actions and supports processes like emotional understanding, action prediction, and imitation. Studies in both humans and animals have demonstrated that mirror-like neural responses appear across different species and play a role in shared action understanding and prosocial behavior. Although mirror neurons have been linked to many aspects of social cognition, much less is known about how these neural systems support social synchrony or collaborative learning, especially when individuals must coordinate their actions with a partner.

The goal of this project is to explore whether mirror-neuron-related processes help support synchronized behavior and coordinated learning in both animals and humans. The first experiment will test whether rats that can observe their partner during a lever-pressing task will show more synchronized responding than rats that complete the task without visual access. The second experiment will use EEG hyperscanning to measure brain activity in human partners as they complete tapping and problem-solving tasks alone and then together. It is expected that joint action will increase both behavioral synchrony and neural alignment. These findings will help clarify how mirror neurons contribute to group coordination and social learning.

## Introduction

The mirror neuron system (MNS) has been widely studied as a neural mechanism that links observing an action with internally simulating or understanding that action. Research originally conducted in monkeys showed that certain neurons fired both when the monkey performed an action and when it observed another individual performing the same action (Ferrari et al., 2003). This discovery led researchers to explore whether similar neural systems supported imitation, action understanding, and social behavior in humans. Because these skills were essential for interacting with others, the MNS became an important candidate for explaining social cognition and coordinated behavior.

Early human research demonstrated that mirror-related activity increased during socially meaningful actions. For example, MNS activation grew stronger when individuals viewed actions that carried social relevance compared to those that did not, suggesting that the system was sensitive to interactive cues rather than simple movement patterns (Oberman et al., 2006). Later work used functional connectivity analyses to show that distinct regions within the human MNS communicated with one another during tasks involving emotional and social interpretation (Sadeghi et al., 2022). These findings suggested that the MNS played a broader role in social processing by helping individuals interpret emotions, facial expressions, and social intentions.

Recent studies expanded this understanding by showing that the mirror system interacted with mentalizing networks responsible for predicting others' goals and intentions. When individuals observed unexpected actions, activity increased in both the MNS and mentalizing regions, and these networks displayed stronger connectivity (Mou et al., 2024). Additional research found that live social interaction produced especially strong mirror-system engagement.

During naturalistic face-to-face interaction, individuals showed enhanced MNS activation and increased effective connectivity between mirror-related regions, indicating that real-time interpersonal engagement evoked richer neural mirroring than passive observation (Hsu et al., 2022).

Other human studies used hyperscanning methods to measure real-time neural synchrony across interacting partners. These techniques showed that shared neural rhythms increased during cooperation, joint attention, and coordinated movement, and recent methodological work outlined how EEG hyperscanning can be used to capture these interpersonal neural dynamics (Zamm et al., 2024). This work supported the idea that neural mirroring contributed to interpersonal alignment and could help explain why individuals often synchronize their actions and emotions during group activities. Additional neuroimaging research suggested that the human MNS may serve as a shared neural basis for social cognition more broadly, connecting action understanding with higher-level processes such as empathy and social prediction (Schmidt, 2021).

Animal research has also played a key role in establishing that mirror-like mechanisms are evolutionarily conserved. Studies in common marmosets identified neurons that fired during both action execution and action observation, demonstrating that mirror responses were present even in New World monkey species (Suzuki et al., 2015). Research in macaques found mirror responses not only for object-directed actions but also for communicative gestures such as lip-smacking, showing that mirror mechanisms extended to social communication (Ferrari et al., 2003). More recent work in rats identified affective mirror and anti-mirror neurons in the anterior cingulate cortex and insular cortex, which activated before prosocial helping behaviors (Wu et al., 2022). These findings suggested that animals used mirroring processes to understand others' actions and emotional states, supporting coordinated and empathetic behavior.

Despite extensive research on mirror neurons in both humans and animals, important gaps remained. Much of the existing work examined MNS activity during observation or imitation, but very few studies directly tested whether mirror neurons contributed to social synchrony, which involves the moment-to-moment alignment of behavior, movement, neural activity, or emotional states between individuals. Synchrony is known to support cooperation, bonding, and collaborative learning, yet the neural mechanisms driving these effects have not been fully identified. Only a limited number of studies have connected mirror-system processes to synchronized behavior in real-time joint tasks.

A second gap involved cross-species comparisons. Human and animal studies often used different methods and focused on different levels of analysis, making it difficult to determine whether the same neural mechanisms supported coordinated behavior across species. Without such comparisons, it remained unclear whether social synchrony depended on a general mirror-based mechanism or on species-specific cognitive processes.

Given these gaps, the present research aimed to investigate whether mirror-neuron-related activity supported synchronized behavior and collaborative learning in both animals and humans. This proposal examined how observation influenced behavioral synchrony in rats and whether joint action increased neural and behavioral alignment in human partners.

## The Current Study

The purpose of this project will be to investigate whether mirror-neuron-related processes will contribute to social synchrony and collaborative learning across species. Although previous work has demonstrated that the mirror neuron system plays an important role in action understanding, imitation, and emotional processing, it remains unclear whether these mechanisms will directly support coordinated behavior between interacting individuals. This proposal will address this gap by testing the behavioral and neural effects of joint action in both rats and humans.

The first aim of this study will be to examine whether observational access will enhance synchronized behavior in rats during a shared lever-press task. The second aim will be to determine whether performing tasks with a partner will increase mirror-neuron-related neural activity and behavioral coordination in humans completing rhythmic and problem-solving tasks.

Hypothesis 1: Rats that will be able to observe a partner completing the lever-pressing task will show higher behavioral synchrony than rats without visual access.

Hypothesis 2: Human participants who will complete tasks together will show greater mu-rhythm suppression, stronger inter-brain synchrony, and more coordinated performance compared to when they complete the same tasks alone.

# Methods

## Participants

The animal portion of the study will use adult male and female Sprague–Dawley rats that will be bred and housed in the university's accredited animal facility. Rats will be between 8 and 12 weeks old at the start of training to ensure that all animals will have reached adulthood and will display stable motor performance. They will be housed in pairs when possible to reduce stress, except in cases where separation will be required for experimental reasons. All rats will be maintained on a standard 12:12 hour light–dark cycle and will have free access to food and water except during behavioral testing sessions, which will occur during the light phase. Environmental enrichment will be provided to support psychological well-being. All procedures will be reviewed and approved by the Institutional Animal Care and Use Committee (IACUC), and all guidelines for humane treatment will be strictly followed.

The human portion of the study will recruit healthy adult participants from the university's student population through flyers, online announcements, and subject pool postings. The sample will include approximately 20–25 pairs of adults aged 18–30 years. Right-handed participants will be selected to reduce hemispheric variability in EEG mu-rhythm measures. Individuals will be excluded if they report a history of neurological disorders, psychiatric diagnoses, seizures, or any condition that could interfere with EEG recording. Participants will provide informed consent and will be compensated with course credit or monetary payment. All procedures will be approved by the Institutional Review Board (IRB).

**Design**

The study will follow a dual-experiment design that will allow comparison of mirror-neuron-related synchrony mechanisms across species. Both experiments will use a mixed design in which participants or animals will experience conditions intended to elicit synchrony as well as conditions that will reduce or eliminate opportunities for observation or cooperation.

In the animal experiment, rats will be randomly assigned to one of two conditions: an observation condition, in which a rat will have full visual access to its partner while the partner performs the lever-press task; or a no-observation control condition, in which an opaque barrier will block visual access. The independent variable will be visual access to the partner (observation vs. no observation). The main dependent variable will be behavioral synchrony, measured as the temporal alignment between lever presses produced by the two rats in a pair.

In the human experiment, all participants will complete both conditions using a within-subjects design. The solo condition will require participants to perform rhythmic tapping and a simple cooperative problem-solving task individually, while the joint condition will require partners to perform the same tasks simultaneously while observing each other. The independent variable will be task condition (solo vs. joint). The dependent variables will include mu-rhythm suppression, inter-brain synchrony, tapping accuracy, and behavioral coordination. Because each participant will complete both conditions, order will be counterbalanced to avoid sequence effects.

## Procedures

### Animal

Rats will undergo a shaping and training phase before experimental testing. During training sessions, each rat will learn to press a lever to receive a small food reward (e.g., sucrose pellet). A shaping procedure will be used to gradually train the lever-pressing behavior, beginning with rewarding the rat for approaching the lever and progressing until the rat performs consistent, reliable presses. Training will continue until both rats in a pair achieve a performance criterion of at least 30 lever presses within a 20-minute session.

Testing will occur in a custom two-chamber apparatus consisting of two operant chambers separated either by a transparent Plexiglas barrier (observation condition) or by an opaque panel (no-observation condition). For each trial, one rat (the “demonstrator”) will perform the lever-pressing task while the other rat (the “observer”) watches or is prevented from watching. After the demonstrator completes a series of presses, the observer rat will be allowed to perform the same task. Video cameras mounted above the chambers will record all sessions.

Behavioral synchrony will be operationalized as the temporal proximity of lever presses between partners across a series of repeated trials. Additional variables such as latency to begin pressing, total number of presses, and accuracy of presses will be recorded. To ensure that synchrony will not be driven by auditory cues, white-noise playback will be used in the testing room. Rats will complete multiple sessions across several days to provide stable performance measures.

**Human**

Human participants will complete two tasks: a rhythmic tapping task and a cooperative problem-solving task. In the tapping task, individuals will tap a response key at a comfortable self-generated rhythm in the solo condition. In the joint condition, pairs will attempt to align their tapping rhythms with one another without verbal communication. In the cooperative task, participants will complete a simple puzzle or decision-making challenge individually and then jointly with their partner. These tasks will be selected to require mutual observation and interaction while remaining simple enough to avoid excessive EEG artifact.

During both the solo and joint conditions, EEG hyperscanning will be used to record neural activity from both participants simultaneously. A 32-channel EEG system will be used, with electrodes placed according to the international 10–20 system. Mu-rhythm activity (8–13 Hz) will be recorded from central electrodes (e.g., C3, Cz, C4) and will serve as an index of mirror-neuron-system engagement. Inter-brain synchrony will be assessed using coherence, phase-locking values, or other hyperscanning metrics. Participants will be seated at a table that allows clear visual access to their partner during the joint condition while minimizing head movement.

Artifacts such as eye blinks, muscle activity, and electrical noise will be identified and removed using independent component analysis. Trials with excessive movement or noise will be excluded. Behavioral synchrony will be quantified by correlating the timing of taps or task-related actions across partners.

## Results

The animal study will include 10–15 pairs of rats per condition based on prior studies of social learning in rodents. Behavioral synchrony will be compared between the two groups using an independent-samples t-test. If performance changes across sessions, a mixed ANOVA with Condition as a between-subjects factor and Session as a within-subjects factor will be conducted. Effect sizes will be reported using Hedges' g, and assumptions of homogeneity and normality will be assessed.

The human sample will include 20–25 pairs of participants. Mu-rhythm suppression and behavioral synchrony will be compared between the solo and joint conditions using paired t-tests. False discovery rate (FDR) correction will be applied to control for multiple comparisons across EEG electrodes and dependent variables. Correlations between neural synchrony and behavioral synchrony will be examined to determine whether greater mirror-system activation will be associated with improved coordination. Hedges' g_av will be used as the effect-size metric for within-subject comparisons, and confidence intervals will be reported for all major analyses.

## Discussion

If the hypotheses are correct, the results will show that mirror-neuron-related processes will contribute to both behavioral and neural synchrony in animals and humans. In the rat experiment, it is expected that rats with visual access to their partner will press the lever in closer temporal alignment than rats in the no-observation condition. Their responses will likely cluster more tightly across repeated trials, and they may also show faster task acquisition due to observational learning. These findings will indicate that the opportunity to observe another individual's actions will enhance the coordination of behavior, which is consistent with the idea that mirroring mechanisms will support synchrony.

In the human study, it is expected that participants will show greater mu-rhythm suppression during the joint condition compared to the solo condition. This pattern will be interpreted as elevated mirror-neuron-system engagement during live social interaction. Human pairs are also expected to demonstrate higher inter-brain synchrony, particularly in sensorimotor and frontal regions associated with shared action understanding. Behaviorally, partners will likely tap with more consistent timing and perform the cooperative problem-solving tasks more efficiently when working together. A positive correlation between neural synchrony and behavioral synchrony will further support the hypothesis that mirror-neuron-related processes will facilitate coordinated social behavior.

If these predictions are supported, the results will provide evidence that mirror-related neural activity will play a functional role in social synchrony across species. Demonstrating similar patterns in rats and humans will strengthen the argument that the basic mechanisms underlying coordinated social behavior are evolutionarily conserved. These findings will contribute to neuroscience by clarifying how shared neural representations may support group learning, cooperation, and real-time interpersonal coordination. Overall, the project will advance understanding of the neural basis of social interaction and provide a framework for future investigations into collaborative cognition.

**Limitations and Future Directions**

One potential challenge in animal study will be that rats may synchronize their behavior due to auditory cues rather than visual observation. To address this, white-noise masking will be used, but if sound continues to influence performance, an additional condition using double-barrier controls or vibration-isolated chambers will be implemented. It is also possible that some rats will not engage socially or will fail to reach the lever-pressing criterion. In this case, additional training sessions or increased pair numbers may be required to maintain statistical power.

In the human study, a major limitation will be the susceptibility of EEG hyperscanning to motion artifacts, especially during interactive tasks. If excessive noise occurs, constraints on movement or alternative task designs that minimize physical motion will be implemented. Another potential issue will be variability in the degree of comfort or familiarity between partners, which may influence synchrony. To reduce this bias, participants will be randomly paired and will complete a brief acclimation period before recording. If neural synchrony does not increase during joint action, alternative interpretations may include the possibility that higher-order cognitive processes, rather than mirror mechanisms, will drive coordination. In that case, adding additional measures such as fNIRS or EMG could help clarify the contribution of different systems.

**Appendix:**

**Figure 1:**

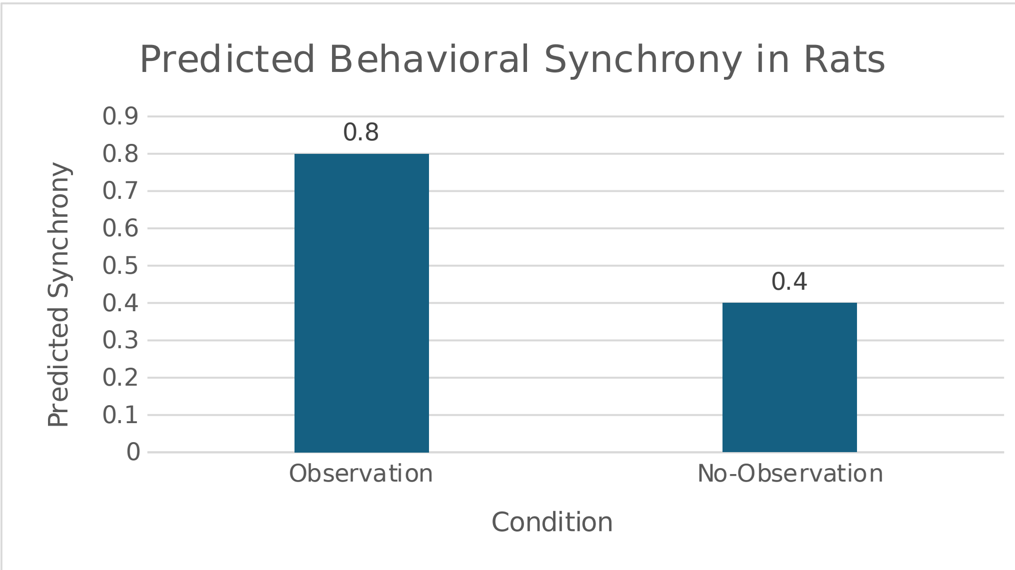


**Figure 1.** *Predicted behavioral synchrony in the rat lever-press task. Rats in the observation condition are expected to show higher synchrony scores than rats in the no-observation condition. These hypothetical values represent the expected direction of results and illustrate the prediction that visual access to a partner will enhance coordinated behavior.*

**Figure 2:**

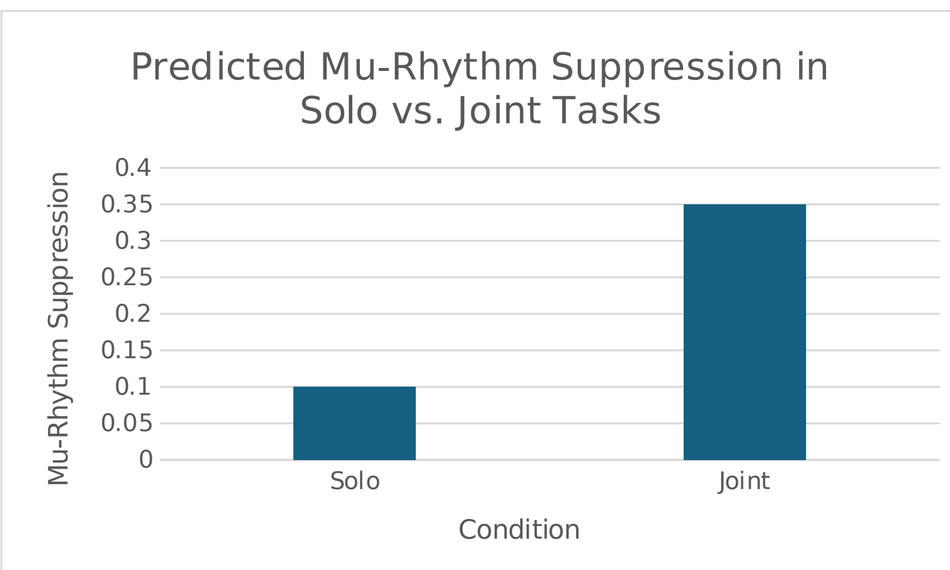

**Figure 2.** *Predicted mu-rhythm suppression during the human joint-action task. Participants are expected to show greater mu suppression when performing the task with a partner compared to when performing alone. These hypothetical data illustrate the expected effect that joint action will increase mirror-neuron-related neural activity.*

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